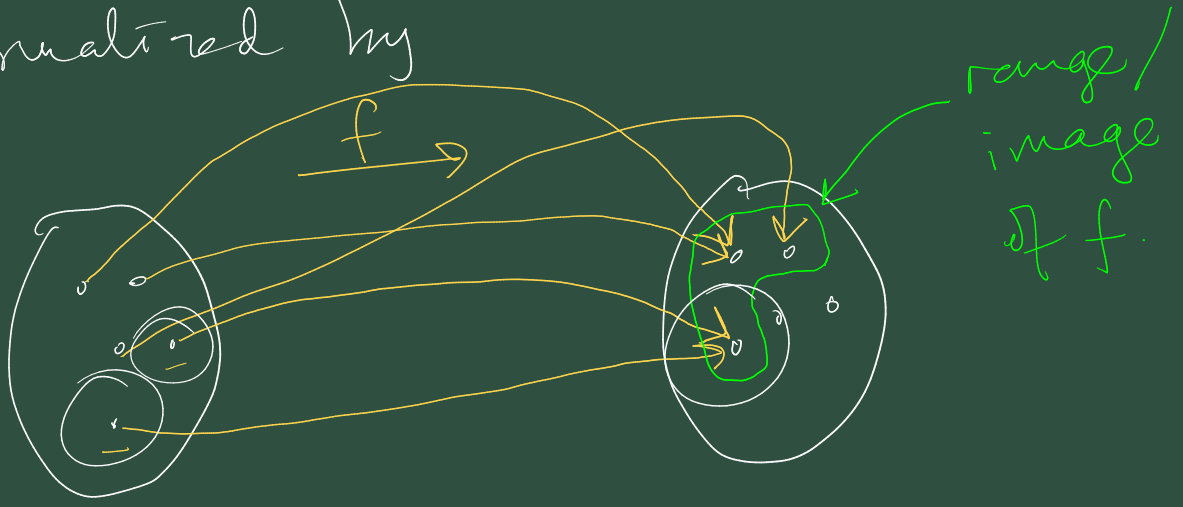
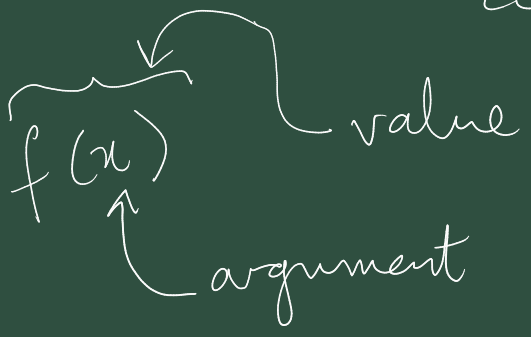
Chap 3.

Some terminology, notation for functions.

especially $f : \mathbb{R} \to \mathbb{R}$.

A function $f : X \to Y$ can be visualized by





$g : U \to V$

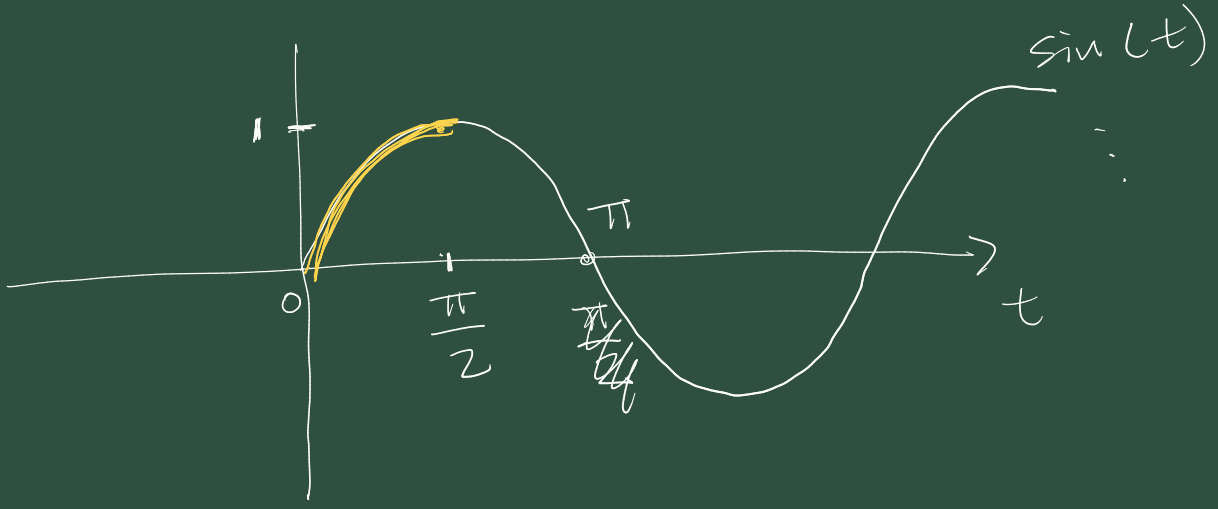
g is injective if for all arguments u_1, u_2 if $g(u_1) = g(u_2)$ then $u_1 = u_2$.

alternatively in the arrow diagram. any element of the co-domain has

at most arrow arriving at it

$g: U \to V$ is surjective

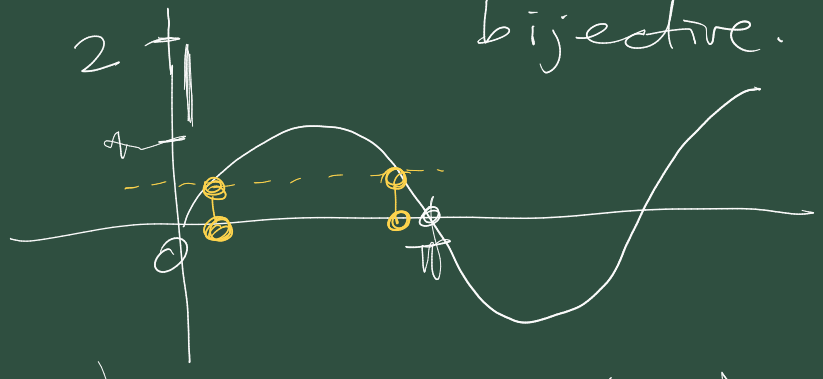
if every element of V has at least arrow arriving at it.



$\sin: (0, \pi/2) \to (0,1)$

is injective and surjective

bijective.

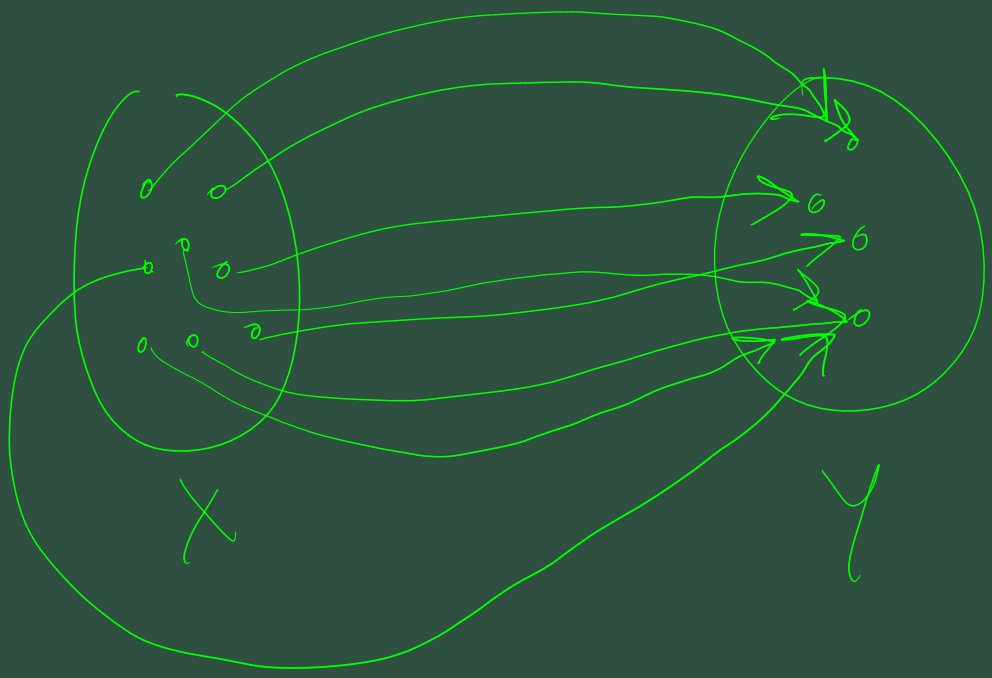


$\sin: (0, \pi) \to (0, 2)$

is not surjective as no ~~value~~ number above 1 occurs as a value.
and is not injective as each value in $(0,1)$ occurs twice over $(0,\pi)$.

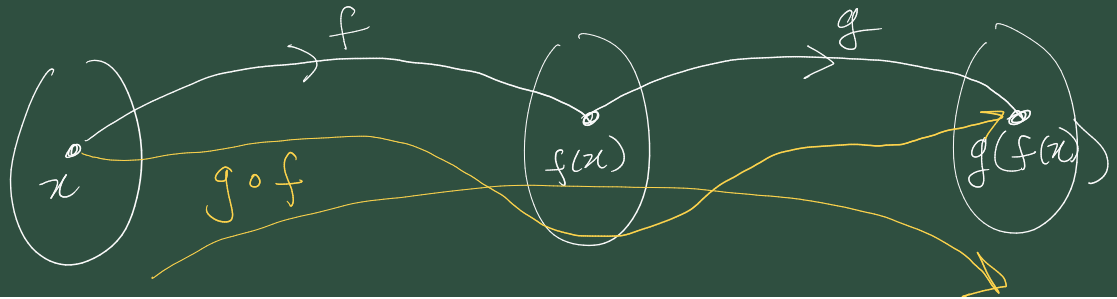
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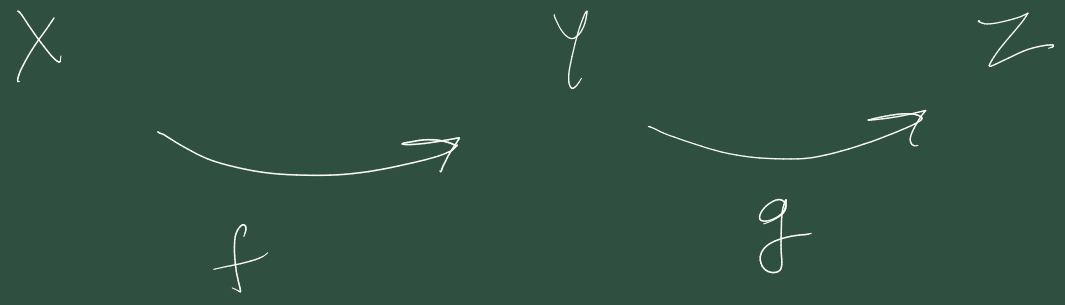
surjective picture:



$f: X \to Y$, where $X, Y \subseteq \mathbb{R}$

i.e. X, Y subsets of the real numbers.





$\mathbb{R} \to \mathbb{R}$

Ex 3.2.7 $g, h, r : \mathbb{R} \to \mathbb{R}$.

3. $g(x) = 2x - 7$, $h(x) = x^2 - 1$

$r(x) = e^{2x}$

Find a simplified formula for

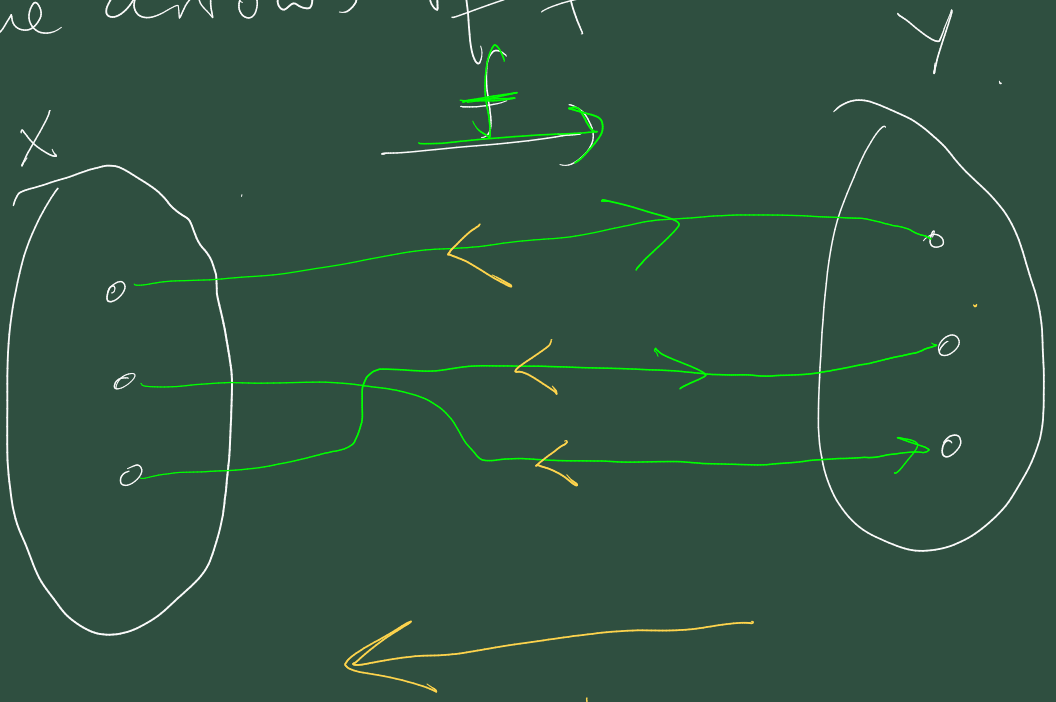
$g \circ h \circ r : \mathbb{R} \longrightarrow \mathbb{R}$

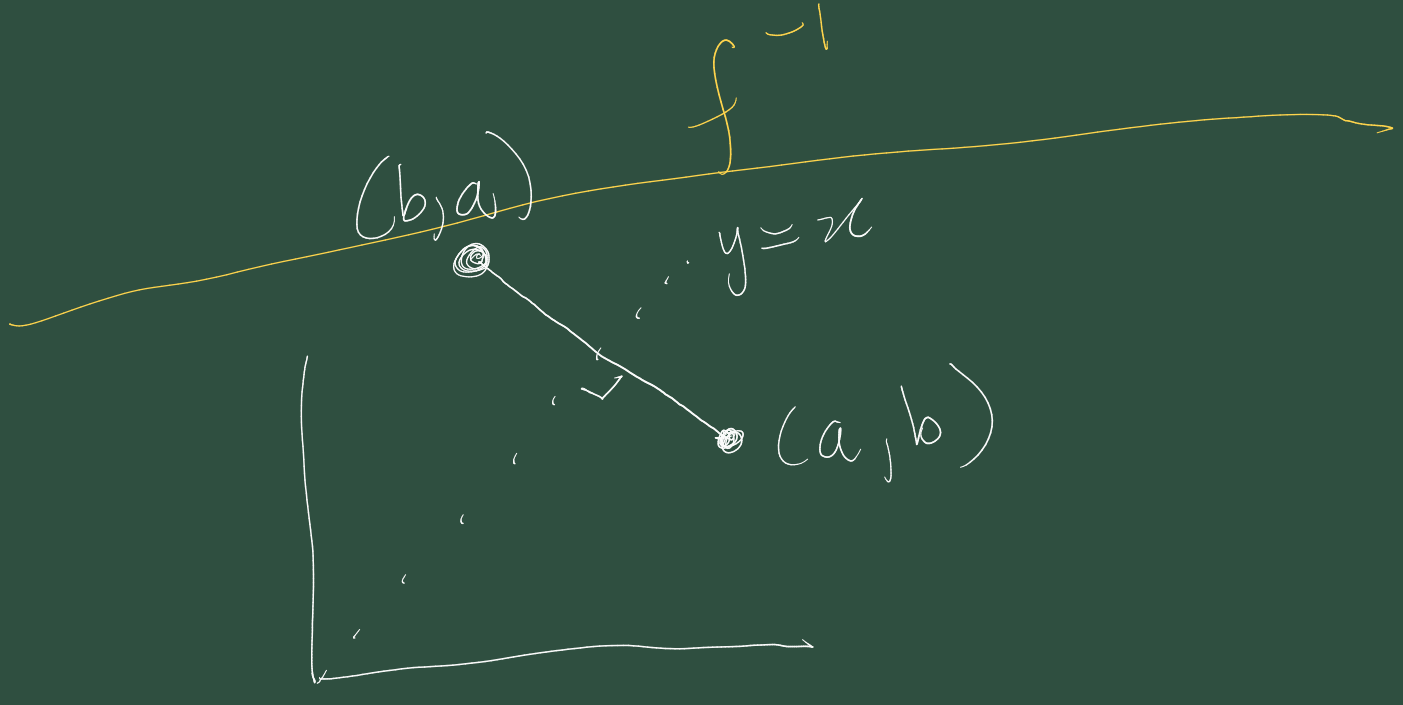
$$(g \circ h \circ r)(x)$$
$$= g(h(r(x)))$$
$$= g(h(e^{2x}))$$
$$= g((e^{2x})^2 - 1)$$
$$= g(e^{4x} - 1)$$

$$= 2(e^{4x} - 1) - 7.$$

$$= 2e^{4x} - 9$$

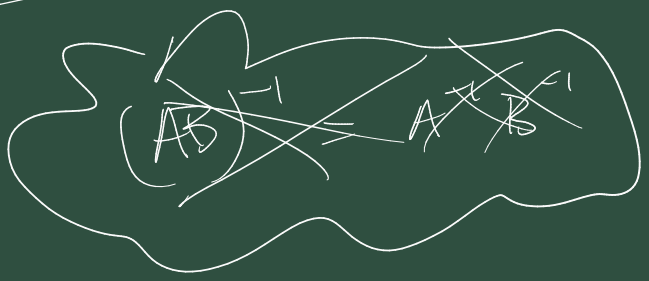
In arrow the inverse function f^{-1} is obtained by reversing all the arrows of f.





In Linear algebra you've seen

$$(AB)^{-1} = B^{-1}A^{-1}$$



3.4 Exercises

Q2.

Consider $f(x) = \left(3e^{2x^2}\right)^{1/2}$.

Express f as a suitable composition of S, R, Exp, One.

(allow ourselves the arithmetic operations $+, \cdot$, scalar multiplication.

Consider

$$f(x) = R\left(3e^{2x^2}\right)$$
$$= R\left(3\left(e^{2x^2}\right)\right)$$

$$= R(3(Exp(2x^2)))$$

$$= R(3(Exp(2(x^2))))$$

$$= R(3(Exp(2S(x))))$$

We see

$$f = R \circ 3Exp \circ 2S$$

(Continue at 12:05)

Ex 3.3.2

Find the expansion of

$$r(x) = \frac{3x+1}{x^3 - 3x + 2}$$

$$= \frac{3x+1}{(x-1)(x^2+x-2)}$$

$$= \frac{3x+1}{(x-1)(x-1)(x+2)}$$

$$= \frac{3x+1}{(x-1)^2(x+2)}$$

$$= \frac{\alpha}{(x-1)} + \frac{\beta}{(x-1)^2} + \frac{\gamma}{x+2}$$

α, β, γ to be determined

$$\Rightarrow 3x+1 = \alpha(x-1)(x+2) + \beta(x+2) + \gamma(x-1)^2$$

$$\Rightarrow 3x+1 = (\alpha+\gamma)x^2 + (\alpha+\beta-2\gamma)x + (-2\alpha+2\beta+\gamma)$$

General principle: polynomials are equal if and only if corresponding coefficients

$\Rightarrow$

$$\left.\begin{aligned} \alpha+\gamma &= 0 \\ \alpha+\beta-2\gamma &= 3 \\ -2\alpha+2\beta+\gamma &= 1 \end{aligned}\right\} \text{ a system of linear equations.}$$

$$\left(\begin{array}{cccc} 1 & 0 & 1 & 0 \\ 1 & 1 & -2 & 3 \\ -2 & 2 & 1 & 1 \end{array}\right) \quad \text{augmented matrix.}$$

$$\left(\begin{array}{cccc} 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 3 \\ 0 & 2 & 3 & 1 \end{array}\right) \qquad r_2' = r_2 - r_1, \quad r_3' = r_3 + 2r_1$$

$$\left(\begin{array}{cccc} 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 3 \\ 0 & 0 & 9 & -5 \end{array}\right) \qquad r_3' = r_3 - 2r_2$$

$$\left(\begin{array}{cccc} 1 & 0 & 1 & 0 \\ 0 & 1 & -3 & 3 \\ 0 & 0 & 1 & -5/9 \end{array}\right) \qquad r_3' = \frac{1}{9} r_3$$

$$r_1' = r_1 - r_3$$
$$r_2' = r_2 + 3r_3$$

$$\begin{pmatrix} 1 & 0 & 0 & 5/9 \\ 0 & 1 & 0 & 12/9 \\ 0 & 0 & 1 & -5/9 \end{pmatrix} \quad 4/3$$

$\Rightarrow$ $\alpha = 5/9$, $\beta = 12/9$, $\gamma = -5/9$.

So our partial fraction is

$$r(x) = \frac{5/9}{x-1} + \frac{12/9}{(x-1)^2} - \frac{5/9}{x+2}$$

3.3.3.

$$S(x) = \frac{5x}{x^3 - x^2 - x - 2}$$ note that 2 is a root of denom.

$$= \frac{5x}{(x-2)(x^2 + ax^2 + x + bx + 1 + c)}$$

has a negative discriminant

$b^2 - 4ac < 0$

so no real roots, ie is irreducible.

$$= \frac{5x}{(x-2)(x^2+x+1)}$$

$$= \frac{\alpha}{x-2} + \frac{\beta x+\gamma}{x^2+x+1}, \text{ for some } \alpha,\beta,\gamma \in \mathbb{R}$$

Multiplying both sides by the overall denominator

$$\Rightarrow 5x = \alpha(x^2+x+1) + (\beta x+\gamma)(x-2)$$

$$= (\alpha+\beta)x^2 + (\alpha-2\beta+\gamma)x + (\alpha-2\gamma)$$

Equate coefficients from both sides.

$$\Rightarrow \left.\begin{aligned} \alpha+\beta &= 0 \\ \alpha-2\beta+\gamma &= 5 \\ \alpha-2\gamma &= 0 \end{aligned}\right\} \text{ simultaneous linear equations.}$$

$$\Rightarrow \left.\begin{aligned} \beta &= -\alpha \\ \gamma &= \frac{\alpha}{2} \\ \alpha+2\alpha+\frac{\alpha}{2} &= 5 \end{aligned}\right\}$$

$$\Rightarrow 3\alpha + \frac{\alpha}{2} = 5$$

$$\Rightarrow 6\alpha+\alpha = 10$$

$$\Rightarrow \alpha = 10/7, \ \beta = -10/7, \ \gamma = \frac{10}{14} = \frac{5}{7}.$$

So our partial fraction expansion is

$$S(x) = \frac{10/7}{x-2} + \frac{-10/7\,x + 5/7}{x^2 + x + 1}$$